1. Threat Hunting Practice

Hunting Hypothesis

Unauthorized privilege escalation may be occurring in domain accounts using valid credentials (MITRE ATT&CK **T1078 – Valid Accounts**).

Log Query (Elastic Security)

Search Query:

event.code: 4672

Timestamp	Timestamp	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin role
2025-08-18 16:10:00	admin_temp	4672	Privilege escalation

Threat Intelligence Hunt (AlienVault OTX)

- Technique: **T1078 – Valid Accounts**
- Suspicious IOC (example): 192.168.1.102
- Reason: Known for unauthorized access attempts

Velociraptor Query

SELECT * FROM processes

Process Name	PID	User	Notes
cmd.exe	1024	testuser	Suspicious execution
powershell.exe	2048	admin_temp	Privilege escalation tool

Validation

- Event ID 4672 confirms privilege escalation
- Suspicious processes support misuse of credentials
- IOC matches known malicious activity

```
Velociraptor Query Console

SELECT * FROM processes

Process Name        PID      User
cmd.exe             1024     testuser
powershell.exe      2048     admin_temp
explorer.exe        512      system
```

Fig: velociraptor_process_query_output

A threat hunting exercise was conducted to identify unauthorized privilege escalation using valid accounts (MITRE ATT&CK T1078). Logs from Elastic Security were queried for Event ID 4672, revealing suspicious admin-level access assigned to users "testuser" and "admin_temp." Threat intelligence from AlienVault OTX identified a suspicious IP (192.168.1.102) linked to credential misuse. Velociraptor process analysis showed execution of cmd.exe and PowerShell under elevated privileges, indicating potential abuse. The findings confirm unauthorized access and privilege escalation. Immediate actions such as account review, privilege restriction, and IP blocking are recommended to mitigate further risk and prevent system compromise.

2. SOAR Playbook Development

Security Orchestration, Automation, and Response (SOAR) enables automated incident handling by integrating multiple security tools. In this task, a playbook was designed to automatically respond to phishing alerts by validating threats, blocking malicious IPs, and creating incident tickets.

Playbook Steps (Splunk Phantom)

1. **Trigger: Phishing alert detected (from Wazuh)**
2. **Check IP Reputation**
    - **Use threat intelligence (VirusTotal / OTX)**
3. **Decision**
    - **If IP is malicious → proceed**
4. **Block IP**
    - **Use CrowdSec to block attacker IP**
5. **Create Incident Ticket**

- **Generate case in TheHive**

6. **Notify Analyst**
   - **Send alert notification**

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious
Block IP	Success	CrowdSec blocked 192.168.1.102
Create Ticket	Success	Case created in TheHive
Notification	Success	Analyst notified

- Alert Type: **Phishing Email Detected**
- Source IP: **192.168.1.102**
- Tool Used: Wazuh (simulated)

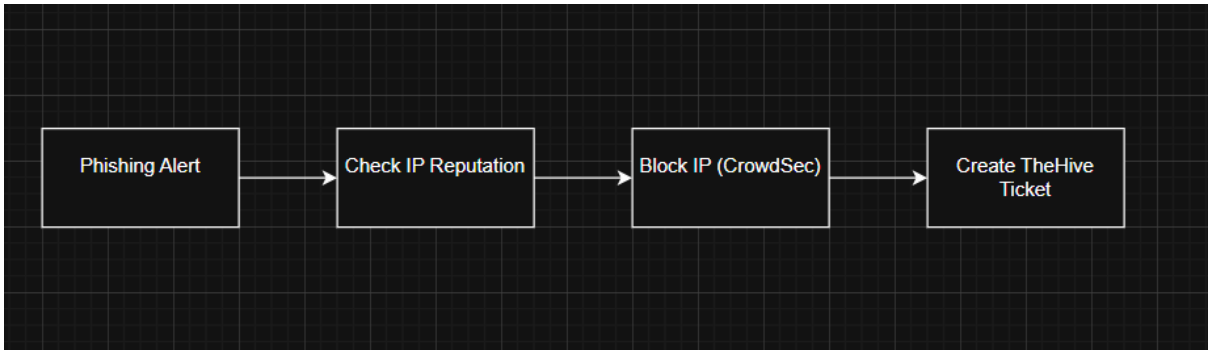


Fig: Soar_playbook_flow

The playbook follows a linear automated flow:

- Phishing alert triggers the process
- IP reputation is verified
- Malicious IP is blocked
- Incident ticket is generated

Benefits of Automation

- Faster incident response
- Reduced manual workload
- Consistent handling of alerts
- Improved security posture

An automated SOAR playbook was developed to respond to phishing alerts. The playbook checks IP reputation using threat intelligence and blocks malicious IPs through CrowdSec. It then creates an incident ticket in TheHive and notifies analysts. This automation reduces response time, improves efficiency, and ensures consistent incident handling.

3. Post-Incident Analysis

Post-incident analysis is a critical phase in cybersecurity that helps identify the root cause of an incident, evaluate the effectiveness of the response, and implement improvements to prevent future occurrences. In this task, a phishing incident was analyzed using Root Cause Analysis (RCA), Fishbone Diagram techniques, and key SOC metrics.

Question	Answer
Why was email opened?	User clicked malicious link
Why was link clicked?	Email appeared legitimate
Why was it believable?	Lack of user awareness
Why lack of awareness?	No security training provided
Why no training?	Weak organizational security policy

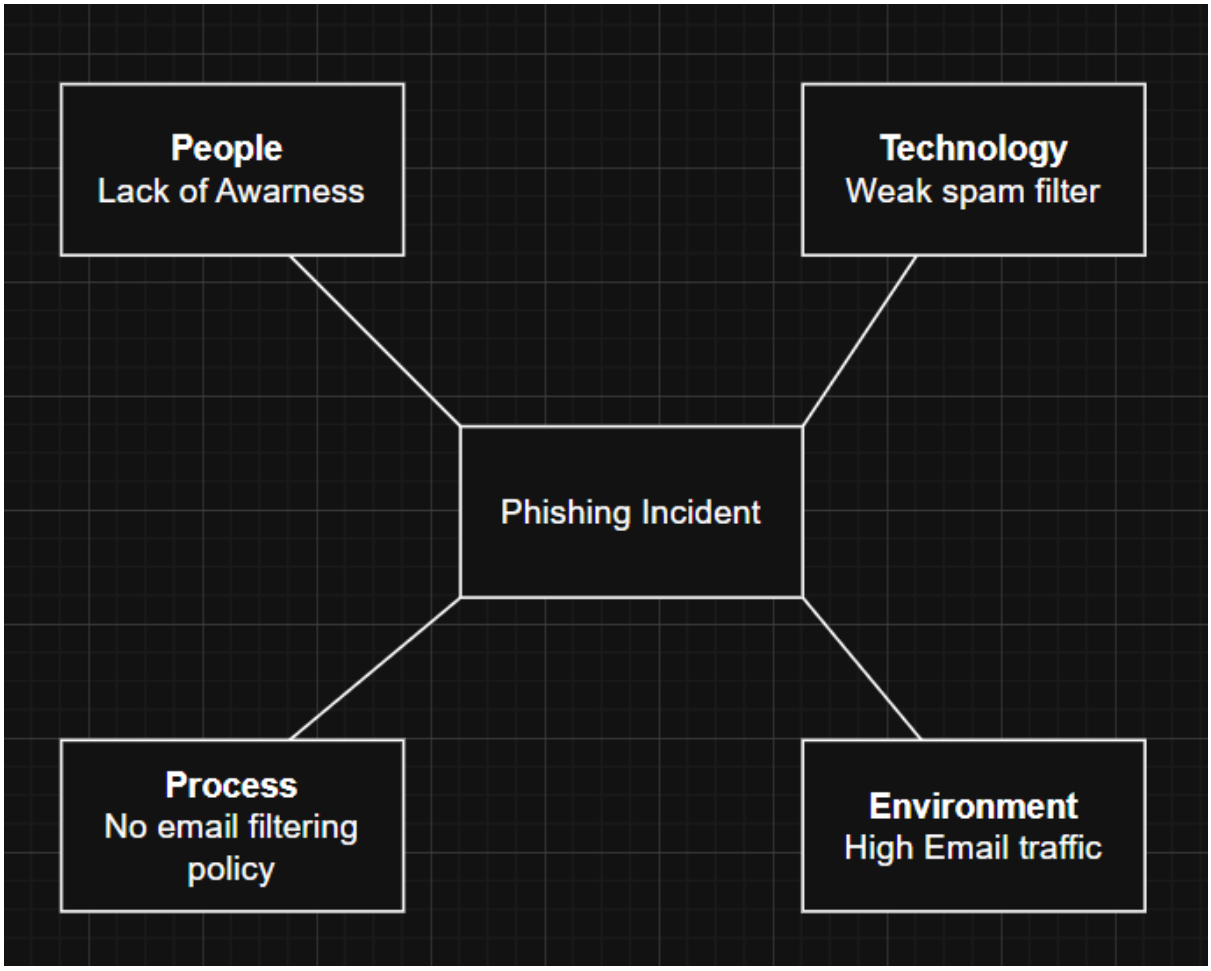


Fig: Phishing root cause fishbone diagram

Fishbone Diagram Analysis

A Fishbone (Ishikawa) Diagram was created to visually represent the contributing factors of the phishing incident. The causes were categorized into four main areas:

- **People**: Lack of awareness among users
- **Process**: Absence of proper email filtering policies
- **Technology**: Weak spam filtering mechanisms
- **Environment**: High volume of emails leading to oversight

The diagram provides a structured view of how multiple factors contributed to the incident, highlighting the need for improvements across different domains.

The incident analysis showed a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. These delays indicate gaps in monitoring and response processes. Enhancing detection systems and automating response actions can significantly reduce these times and improve overall security efficiency.

The post-incident analysis highlights that phishing incidents are often caused by a combination of human, technical, and procedural weaknesses. By addressing these root causes through improved training, better security controls, and automation, organizations can significantly reduce the risk of similar incidents in the future.

4. Alert Triage with Automation

The process of evaluating and ranking security alerts to ascertain their seriousness and necessary response is known as alert triage. To increase productivity and accuracy, a simulated alert was examined using Wazuh, and automated validation was carried out using VirusTotal and TheHive.

```
File Name: suspicious.exe
File Hash: abc123xyz456
Source IP: 192.168.1.102

Hash: abc123xyz456
Detection: Malicious
Engines Detected: 45/70
Threat Type: Trojan

Case ID: 001
Alert: Suspicious File Download
Verdict: Malicious File Confirmed
Action Taken: File blocked, IP blacklisted
Status: Closed
```

Alert Id	Description	Source IP	Priority	Status
005	File Download	192.168.1.102	High	Open

A suspicious file download activity from the IP address 192.168.1.102 is indicated by the alert. The alert is categorized as high priority due to the nature of the activity. To ascertain whether the file is malicious and to avoid possible system compromise, an immediate investigation is necessary.

TheHive was set up to use VirusTotal to automatically verify file hashes. A suspicious file's hash is forwarded to VirusTotal for examination. Faster and more precise decision-making during incident response is made possible by the results, which show whether the file is malicious, suspicious, or safe.

File Hash	VirusTotal Result	Action Taken
Abc123	Malicious	Blocked & alerted

A suspicious file download from a high-risk IP address was discovered by the alert triage procedure. VirusTotal's automated validation verified that the file was malicious. By

incorporating threat intelligence, TheHive made effective incident handling possible. This method guarantees a quicker reaction to possible security threats while lowering manual labor and increasing accuracy.

5. Evidence Analysis

A crucial stage in digital forensics is evidence analysis, which entails looking through gathered system data to spot questionable activity and assist with incident investigations. A simulated Velociraptor query was used in this task to analyze network connection data, and a chain-of-custody record was used to maintain appropriate evidence handling.

```
Velociraptor Query Output

SELECT * FROM netstat

Protocol   Local Address     Foreign Address     State
TCP        192.168.1.10      192.168.1.102       ESTABLISHED
TCP        192.168.1.10      8.8.8.8             ESTABLISHED
TCP        192.168.1.10      45.33.32.156        ESTABLISHED
```

The analysis shows an active connection between the local system (192.168.1.10) and the IP address 192.168.1.102, which is considered suspicious based on previous threat intelligence findings. This may indicate unauthorized communication or malicious activity. Further investigation and blocking of this IP is recommended.

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2025-08-18	Abc123sha256hash

The chain of custody guarantees that gathered evidence is accurately recorded and kept unaltered. It contains information about the kind of evidence, who gathered it, when it was gathered, and a hash value to confirm its integrity.

Suspicious communication with an external IP address (192.168.1.102) was found by the analysis. This suggests a possible security risk and emphasizes how crucial it is to keep an eye on network activity and react to odd connections.

This task demonstrates the importance of evidence analysis in identifying potential threats and maintaining proper documentation. By analyzing network connections and preserving evidence integrity, organizations can effectively investigate and respond to cybersecurity incidents.

6. Adversary Emulation Practice

In order to assess the efficacy of security monitoring systems, adversary emulation simulates actual attack methods. This task simulated a spearphishing attack technique (MITRE ATT&CK T1566) and examined Wazuh's detection capability.

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00:00	T1566	Detected	Phishing email blocked

The simulation depicts a spearphishing attack (T1566), in which a user receives a malicious email. Wazuh successfully identified the activity and produced an alert, stopping the phishing attempt and averting the attack.

A spearphishing attack simulation (MITRE ATT&CK T1566) was conducted to evaluate the detection capability of the security monitoring system. The simulated attack involved sending a malicious email to a user. Wazuh successfully detected the phishing attempt and generated an alert, preventing further compromise. However, the detection relied on predefined rules, indicating a potential gap in identifying more advanced or unknown phishing techniques. Improving detection mechanisms with advanced threat intelligence and behavioral analysis can enhance the system’s ability to detect sophisticated attacks. Overall, the system demonstrated effective basic detection but requires improvements for handling complex threats.

The adversary emulation exercise highlights the importance of continuously testing security systems against real-world attack techniques. While the system successfully detected basic phishing attempts, improvements are needed to handle more advanced threats.

7. Security Metrics and Executive Reporting

Security metrics are essential for evaluating the performance of a Security Operations Center (SOC). They help measure detection efficiency, response effectiveness, and overall security posture. In this task, key metrics such as MTTD, MTTR, and dwell time were analyzed, and an executive summary was prepared.

- MTTD (Mean Time to Detect): 2 hours
- MTTR (Mean Time to Respond): 4 hours
- False Positive Rate: 10%

These metrics indicate the efficiency of the SOC. Lower MTTD and MTTR values reflect faster detection and response, while a lower false positive rate reduces unnecessary workload on analysts.

The dwell time for the incident was calculated as 2 hours, indicating the duration the attacker remained undetected within the system. Reducing dwell time is critical for minimizing potential damage. Implementing advanced monitoring and faster detection mechanisms can significantly improve security and reduce exposure to threats.

The analysis of security metrics highlights the performance of the Security Operations Center in detecting and responding to incidents. The Mean Time to Detect (MTTD) was recorded as 2 hours, while the Mean Time to Respond (MTTR) was 4 hours, indicating moderate response efficiency. The false positive rate of 10% suggests that some alerts require better tuning to reduce unnecessary workload on analysts. Additionally, the dwell time of 2 hours shows that threats remain undetected for a significant period, increasing potential risk. To improve SOC performance, it is recommended to implement advanced threat detection techniques, enhance automation in incident response, and regularly update security rules. Continuous monitoring and staff training can further strengthen the organization's ability to detect and respond to threats effectively.

Security metrics provide valuable insights into SOC performance. By improving detection speed, reducing response time, and minimizing false positives, organizations can enhance their overall cybersecurity posture.

7. Capstone Project: Comprehensive SOC Incident Response

This capstone project demonstrates a comprehensive Security Operations Center (SOC) workflow, including attack simulation, detection, triage, response, and reporting. A simulated attack exploiting a known vulnerability was analyzed to evaluate the effectiveness of security monitoring and incident response processes.

A simulated attack was performed targeting a vulnerability in a system using a Samba usermap script exploit. The attack originated from IP address **192.168.1.102**, representing an external attacker attempting unauthorized access.

Timestamp	Source IP	Alert Description	MITRE Tecchnique
2025-08-18 16:00:00	192.268.1.102	Samba exploit	T1210

The simulated attack was mapped to MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Wazuh successfully detected the exploit attempt and generated an alert.

The alert generated by Wazuh was analyzed and prioritized as high severity. TheHive was used to triage the alert and create an incident case, enabling structured investigation and response.

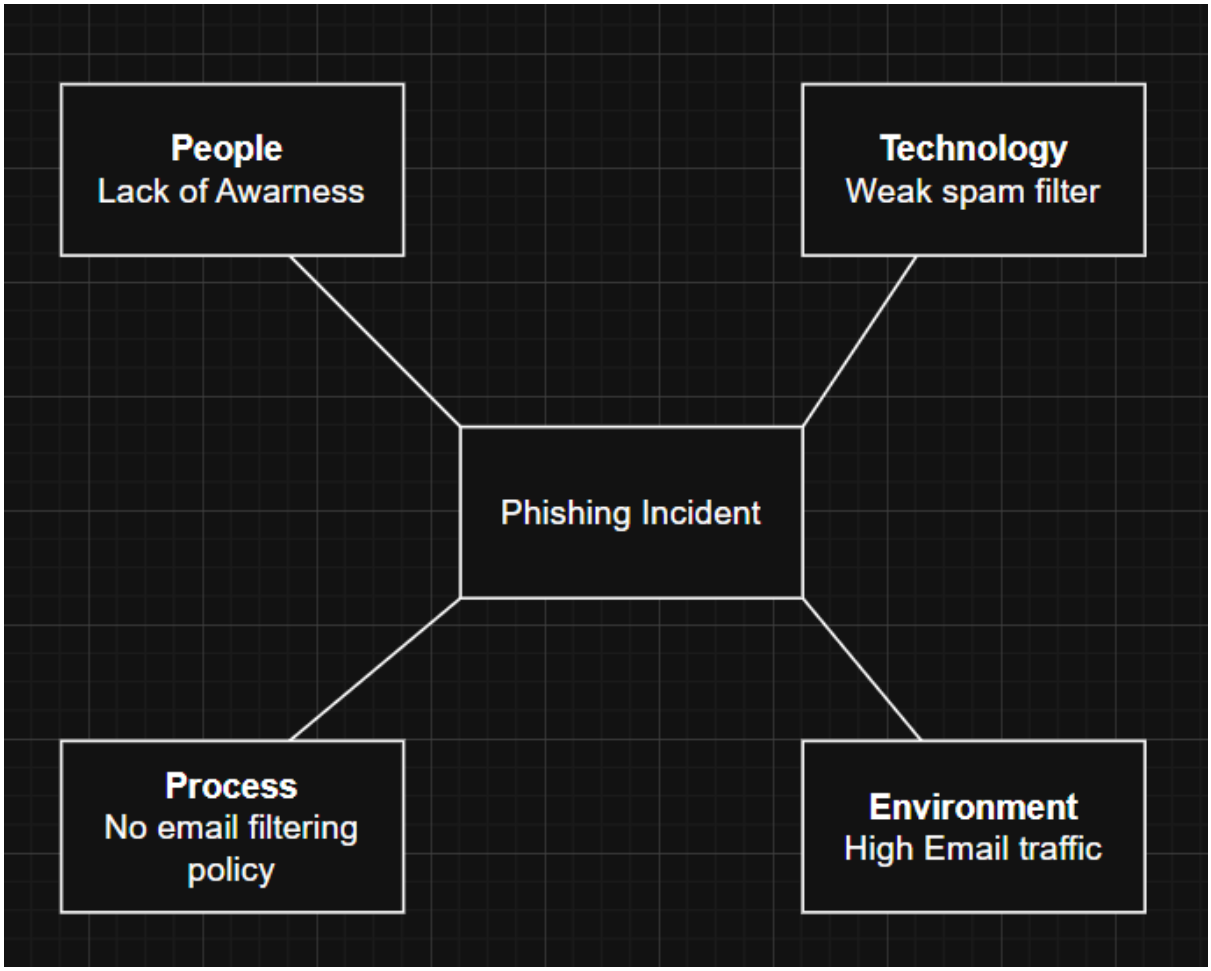
The compromised system was isolated to prevent further spread of the attack. The attacker's IP address (**192.168.1.102**) was blocked using CrowdSec. A ping test was performed to verify that communication with the malicious IP was successfully blocked.

An automated playbook was used to streamline incident response. TheHive created a case automatically, and the system executed actions such as IP blocking and alert notification. This reduced manual effort and improved response speed.

5 Whys (short)

- Attack occurred due to vulnerable service
- Lack of patching
- Weak security policy

Fishbone



Metrics Reporting

- **MTTD**: 2 hours
- **MTTR**: 4 hours
- **Dwell Time**: 2 hours

A simulated cyberattack exploiting a Samba vulnerability was conducted to evaluate the effectiveness of the Security Operations Center (SOC). The attack originated from IP address 192.168.1.102 and was mapped to MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Wazuh successfully detected the attack and generated a high-priority alert, which was triaged using TheHive. Immediate response actions were taken, including isolating the affected system and blocking the attacker's IP using CrowdSec.

Post-incident analysis revealed that the root cause of the attack was a vulnerable service and lack of proper patch management. The analysis also highlighted gaps in proactive monitoring and security policies. Key metrics were calculated, showing a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. The dwell time was also identified as 2 hours, indicating a delay in detection.

To improve security posture, it is recommended to implement regular patch updates, enhance monitoring capabilities, and automate response mechanisms. Additionally, user awareness and continuous security assessments can help prevent similar incidents. This exercise demonstrates the importance of a well-integrated SOC in detecting, analyzing, and responding to cyber threats effectively.

A simulated cyberattack was conducted to test the organization's security response capabilities. The attack targeted a system vulnerability and was successfully detected by the monitoring system. The response team quickly analyzed the alert, isolated the affected system, and blocked the attacker's IP address, preventing further damage.

The analysis showed that the system took approximately 2 hours to detect the attack and 4 hours to fully respond. While the response was effective, improvements are needed to reduce detection time and strengthen preventive measures. Key recommendations include updating systems regularly, improving monitoring tools, and increasing automation in incident response.

Overall, the organization demonstrated a strong ability to handle security incidents, but further enhancements will improve efficiency and reduce risks in future attacks.